Hochschule für Technik und
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Bachelor Thesis

Data Visualization for Augmented Reality Head-Up Displays in Personal Protective Equipment

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1 Introduction

1.1 Motivation

While working in personal protection equipment, it is essential to be able to parse critical information, such as oxygen levels, at a glance. Chemical protective suits in particular reduce peripheral vision and mobility (Murray et al., 2011), making it challenging to operate radio devices and other tools.

A possible solution might be a head-up display (HUD), since it could integrate information into the field of view. Numerous studies confirm an increase in task performance when using a HUD (Chauvin et al., 2023; Kim et al., 2019).

This thesis aims to analyze requirements and current technological issues, as well as propose potential user interface designs for HUDs independent of hardware implementation.

2 Potential Usage Scenarios

Confined Space Entry

Chemical Protection, Firefighters, Cleanrooms

Common Characteristics, especially high mental load + stress distinction to other fields of application diminished agility etc

Vital to not cut off senses if system breaks → optical see-through

3 Biological and Psychological Requirements

vision + focus distance

reaction to color + movement

3.1 Vision Properties

3.2 Psychological Safety

4 Current Technological Limitations

Display-Größe, Anordnung Helm/Atemmaske vs Schutzanzug

o diagram side view CPC

4.1 Background Variation and Contrast

50-70% transparency (Hussain & Park, 2024)

turn down opacity in darker env → like brightness

dark env + torch: harsh lighting makes difference?

2 layers: main info (HUD, doesn't move) + AR (hazard highlighting, moves)

overlap? "border" where highlights disappear before overlapping w/ HUD?

4.2 Movement Tracking

evtl. nur in Ausblick

5 Conceptualization

5.1 Data Management

what data do i need? bio vs hardware?

- heart rate variability - oxygen measurement

how do i get realistic dummy data?

5.2 Data Visualisation

effects of color? monochrome? w/ accents?

would be interesting to see if certain colors would have a calming effect, thus saving oxygen or sth

diagrams vs just a number? mainly number, diagram in special cases like oxygen use

show units? Y, except low on space/minimal version

movement

pop up warning

iconography

traffic light system

5.3 UI Elements

requirements: lowest mental/sensory load possible

Diagrams vs just a number (movement of diagram might be distracting)

language?

positioning elements? some directions seem to be favored for distractability?

FOV diagram w/ angles

do i need to leave space at the sides for peripheral vision? how much?

2 full examples: full/more detailed on CPC glass, minimal version on goggles

each on dark + varied bg

just icons for hazardous atmosphere/environment

6 Visual Prototype

properties of used hardware (probably MetaQuest 3)

development process

7 Conclusion

Appendix

References

Chauvin, C., Said, F., & Langlois, S. (2023). Augmented reality HUD vs. conventional HUD to perform a navigation task in a complex driving situation. *Cognition, Technology & Work*, *25*(2–3), 217–232. https://doi.org/10.1007/s10111-023-00725-7

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